



Code-Layout, Readability and Reusability

Date	02 October 2026
Team ID	SWTID-2026-7991
Project Name	EduGenie: Google Gemini Powered Learning Assistant
Maximum Marks	5 Marks

Code-Layout, Readability and Reusability:

This document evaluates the quality of the EduGenie codebase in terms of structure, readability, and reusability. The implementation uses modular components, clear naming, comments, and reusable services to support maintainability and future extension.

Code Layout Checklist:

S.No	Code Quality Parameter	Description	Followed (Yes / No / Partial)	Remarks
1	Consistent Indentation	Uniform spacing and formatting used throughout the code	Yes	Python/React files follow consistent formatting.
2	Proper File Structure	Files and folders are logically organized	Yes	Frontend, backend, and configuration are separated.
3	Meaningful Variable Names	Variables reflect their purpose clearly	Yes	Descriptive names are used for learning workflows.
4	Function / Method Names	Functions are descriptively named	Yes	Names identify API and learning operations clearly.
5	Code Comments	Comments explain important logic	Yes	Comments/documentation are included where useful.
6	Modular Design	Code is split into reusable functions/modules	Yes	FastAPI routes and learning modules are separated.
7	No Redundant Code	Duplicate or unused code is minimized	Yes	Common logic is kept reusable.
8	Error Handling	Exceptions and errors are handled gracefully	Yes	API validation and error responses are handled.

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Reusable Components / Modules:

S.No	Component / Module Name	Language / Technology	Where Reused	Reusability Level (High / Medium / Low)
1	Gemini Service	Python	Q&A, explanations, summaries, quizzes	High
2	FastAPI API Routes	Python / FastAPI	All frontend learning requests	High
3	Learning Components	React / JavaScript	Question, summary and quiz screens	High
4	Prompt / Quiz Utilities	Python	Quiz and personalized learning flows	High
5	Data Access	SQLite	Session and quiz-result storage	Medium

Overall Code Quality Assessment:

Aspect	Rating (1-5)	Comments
Code Layout & Structure	5	Organized into frontend, backend, API, and service modules.
Readability	5	Clear names and consistent structure support easy reading.
Reusability	5	Shared learning and API components are designed for reuse.
Documentation / Comments	4	Key setup and logic areas are documented.
Overall Score	4.75 / 5	Strong structure with scope for further documentation.